THIS IS HOW VIETNAMESE THINK ABOUT FOOD,

AND LOOK FOR FOOD...

Yo! That bowl of pho we smashed the other day was fire, right?

Yo! hôm trước đá bát phở ngon nhỉ

Nhỉ peak vãi, nóng hổi vừa thổi vừa ăn

Peak AF, isn't it? Piping hot, blowing while eating

...BY MEMORY, BY VIBE, IN SLANG,
AND NOT DESCRIBED ON MAPS' SEARCH BAR.

Personalized Query Auto-Completion for Large-Scale POI Search at Baidu Maps 70:5

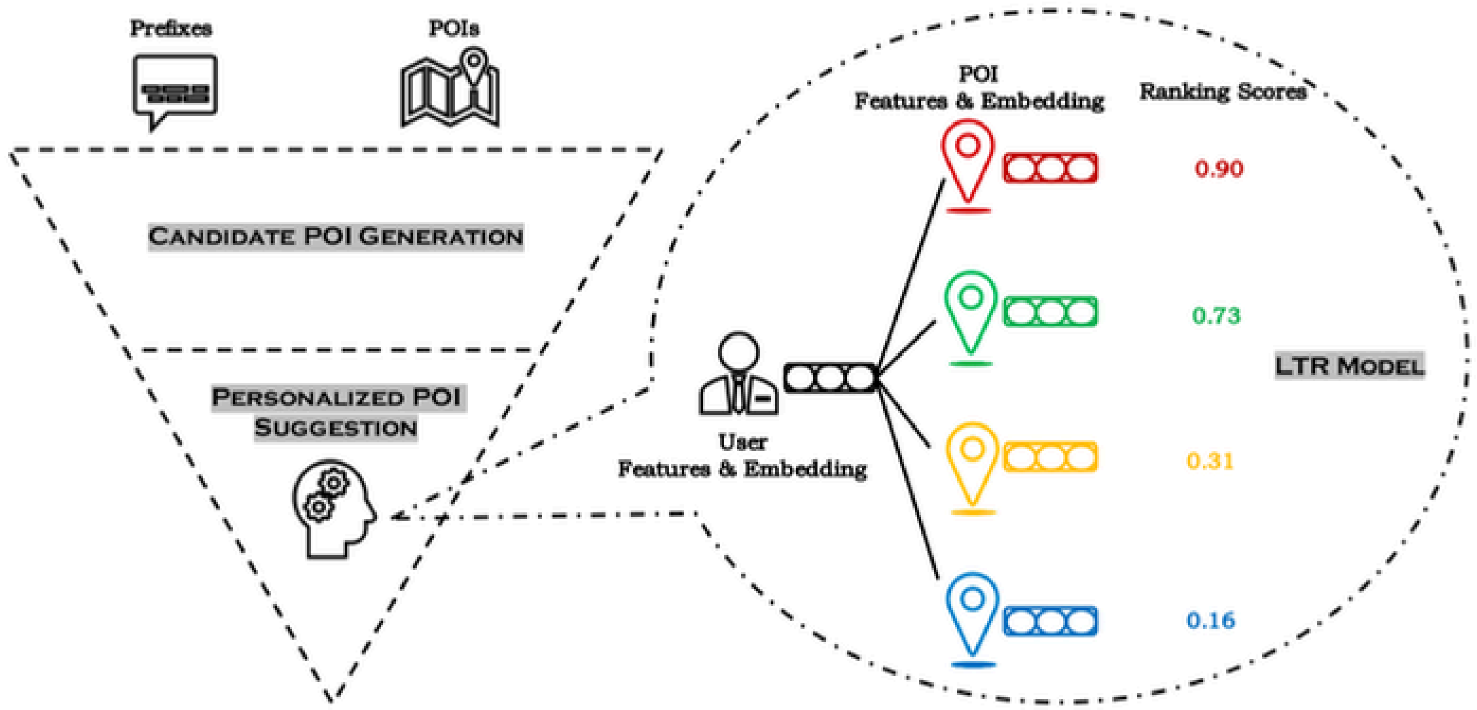


Fig. 3. The framework of the QAC module deployed in the POI search engine of Baidu Maps. Generally speaking, this funnel-shaped framework is composed of two layers: the layer of candidate POI generation on the top and the layer of personalized POI suggestion at the bottom. The top layer (i.e., candidate POI generation) is responsible for filtering out most POIs that are irrelevant to the given prefix. The bottom layer (i.e., personalized POI suggestion) takes hundreds of generated POI candidates as input and learns to rank them in accordance with the features of both users and POIs.

GLOBAL FOOD IE EXIST, BUT FAILING TO LOCALIZE TO OUR CULTURE PREVENTS THEM FROM BEING EFFECTIVELY PERSONALIZED IN VIETNAM.

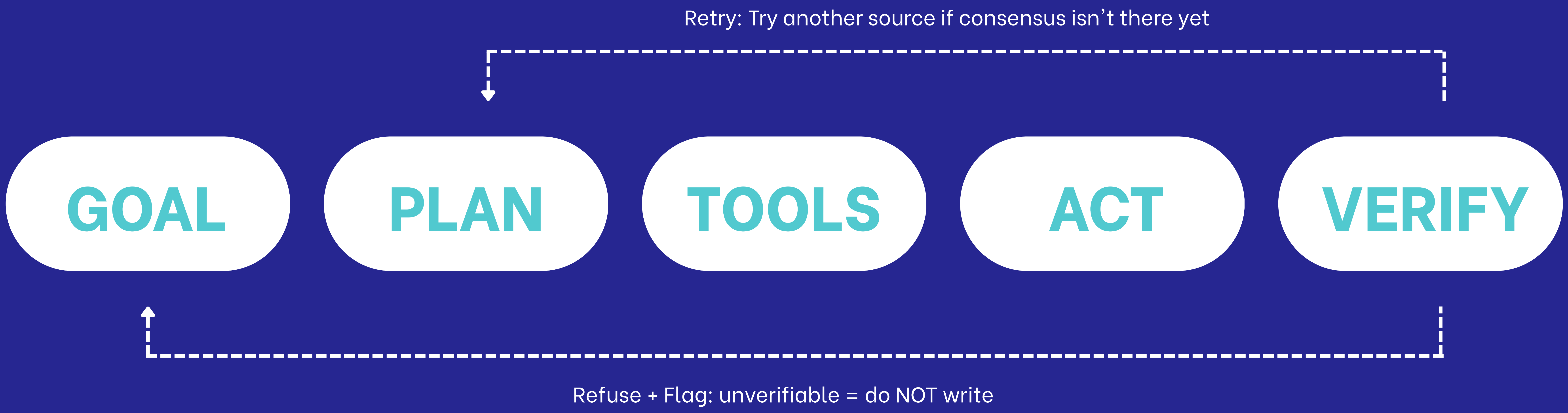
SOLUTION

Reproduce the giants' food-intelligence engines for **individualization**

+

Localized to Vietnamese's context

OUR AGENTIC LOOP



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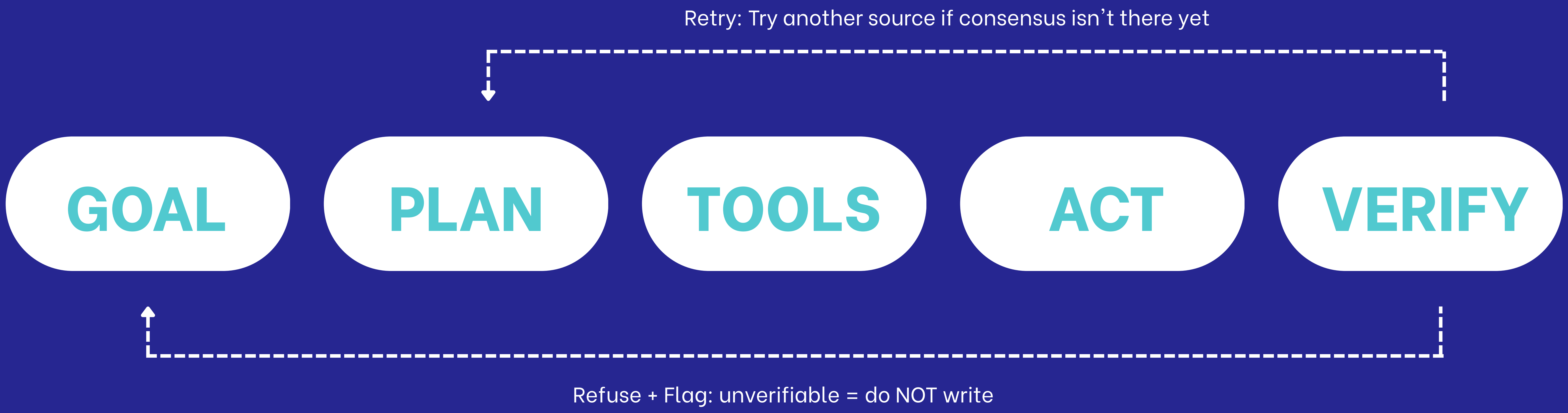


Team: toilaAI

Tasco Maps AI-powered restaurant intelligence

collect, verify, and score POI data – a trusted personalized AI recommendation system for Vietnamese drivers

Appendix: Problem

1

Unspoken context **isn't personalized in real-time**, and **learned by time:**
from parking, what-to-eat-when, to personal preferences

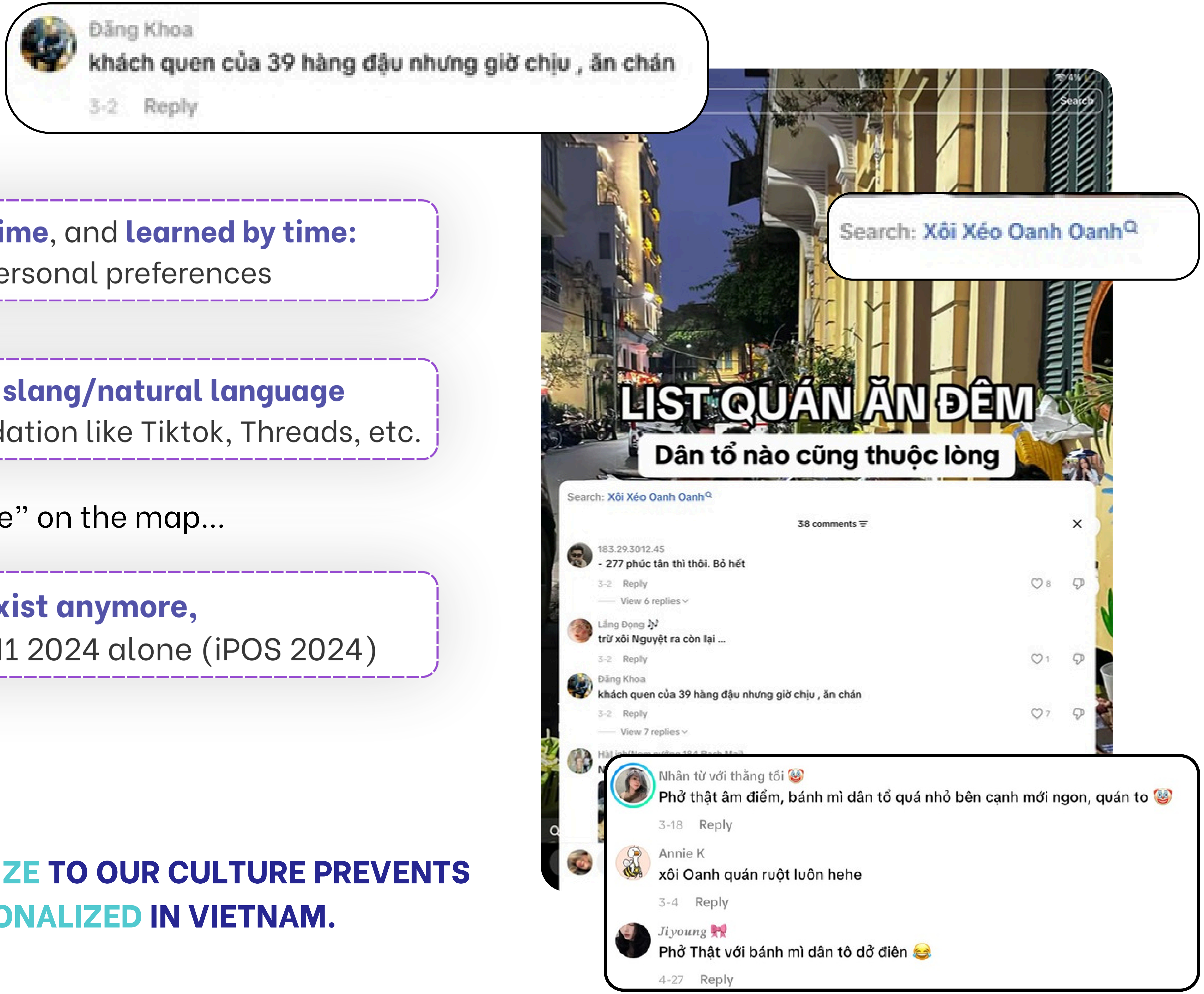
2

Google Map doesn't understand **Vietnamese slang/natural language**
to give diverse, commonly agreed POI recommendation like Tiktok, Threads, etc.

and even when you find "the one" on the map...

3

the place you found **might not exist anymore,**
cause **<u>in Vietnam</u>**, 30,000 outlets closed in H1 2024 alone (iPOS 2024)

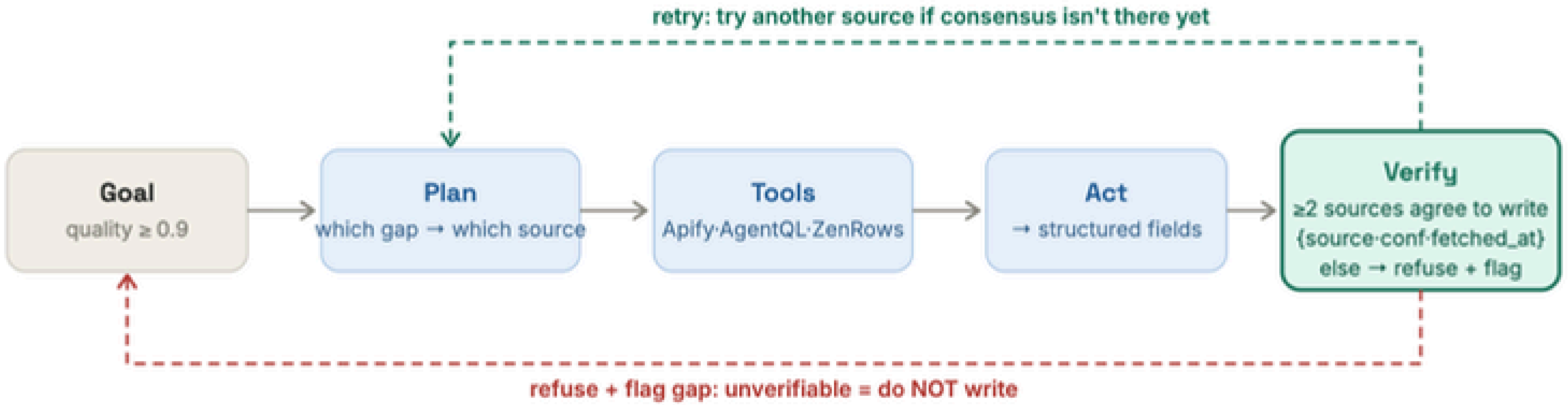


GLOBAL FOOD IE EXIST, BUT FAILING TO LOCALIZE TO OUR CULTURE PREVENTS THEM FROM BEING EFFECTIVELY PERSONALIZED IN VIETNAM.

Appendix: Our agentic loop

The agentic loop: a different plan for every POI

A pipeline runs fixed steps. An agent decides per-POI, and knows when to refuse.



EXAMPLE DECISIONS PER GAP (a different plan per POI):

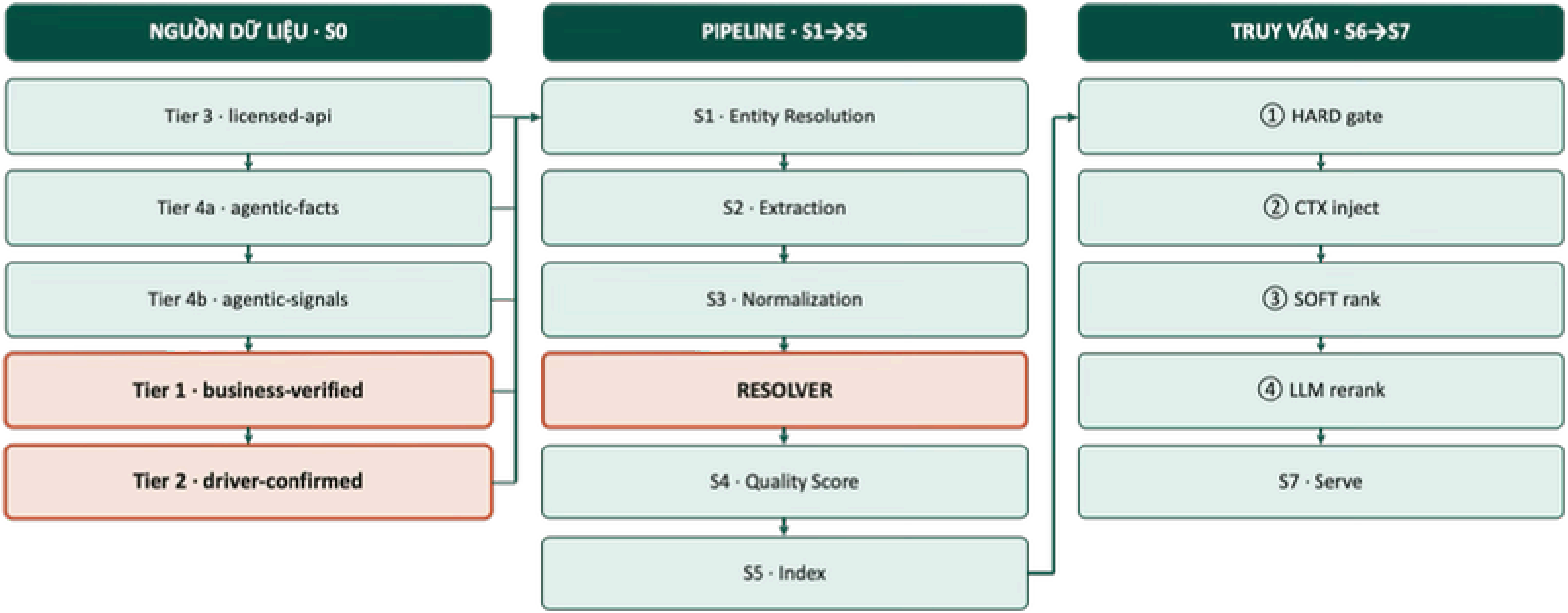
gap: opening hours
→ Google Places actor + FB Page
consensus 2/2 → write, conf 0.80

gap: menu + prices
→ Foody (AgentQL) + website (Tavily)
price parsed by regex, LLM never guesses pric

gap: dietary — only 1 source says "vegetarian"
consensus 1/2 → NOT written
gap flagged, waiting on a second source

The three example boxes are ammo for "why agentic, not a pipeline?": different POI → different plan; and the red box proves it knows how to **refuse**. Scope check: this runs on **real POIs outside the 30-quán benchmark only** (PLAN.md §6.5) and sits in the nice-to-have tier — cut first if the 12-hour build runs short (§10). Don't present it as guaranteed-live without checking the day-of build status.

Appendix: Core system flow

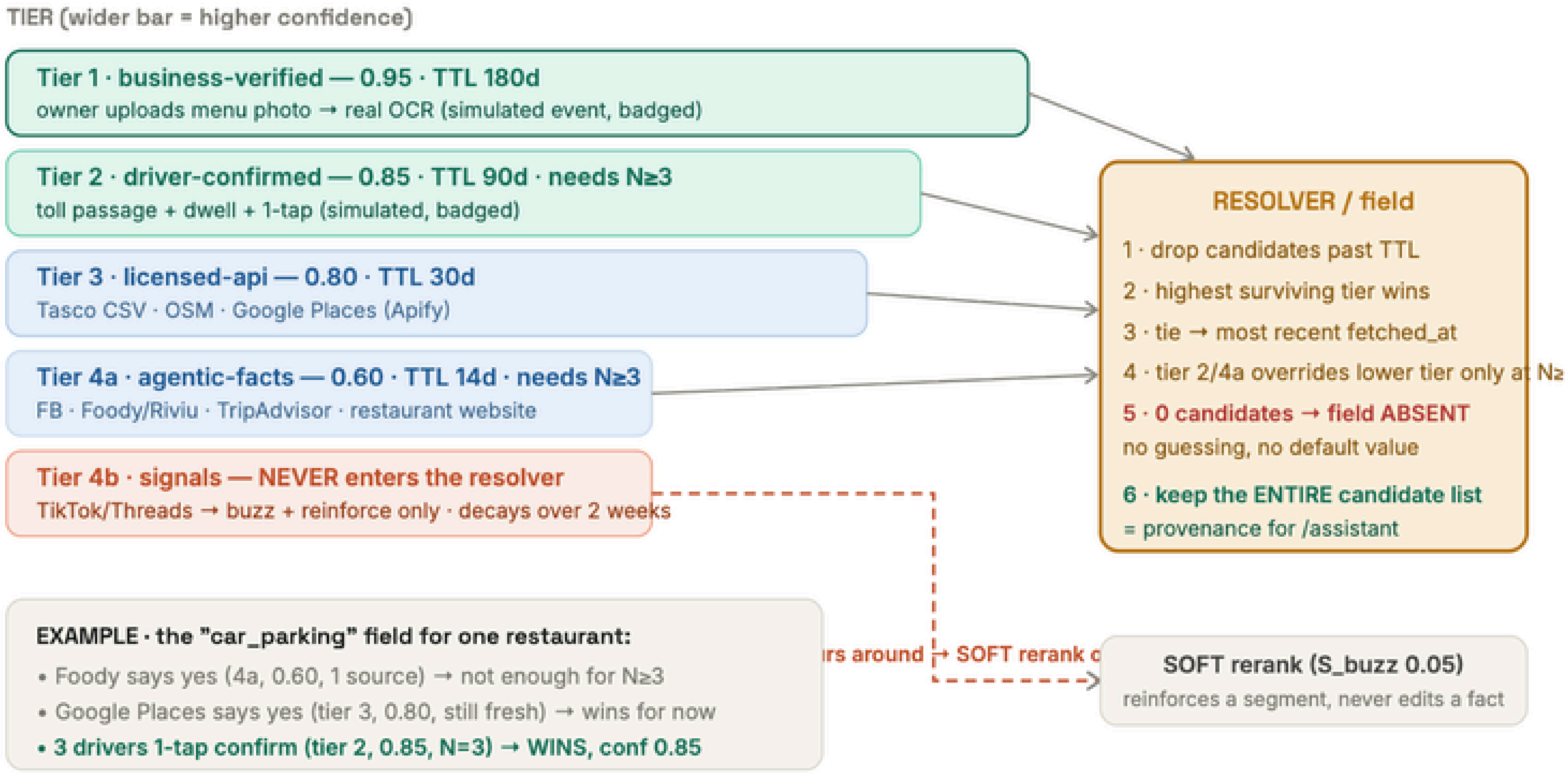


Appendix: Source Tier

G4 · APPENDIX A3 · TRUST MODEL

5-tier cascade: a fact is only written if it wins a contest between sources

Every field is an auction between candidates. Highest tier wins · lower tiers need N≥3 consensus · signals never compete.



This diagram answers "data quality of scraped facts": **no single scraped source ever becomes a published fact alone**. The duel example shows Tasco's driver data outranking every public source — that's the moat living inside the mechanism, not just on a slide.

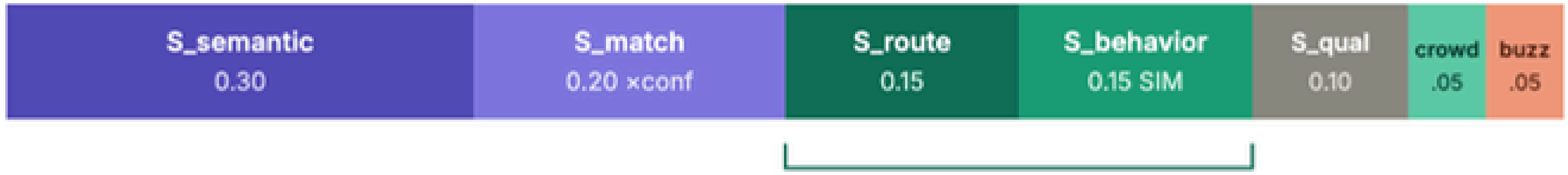
Appendix: Ranking Model

G5 · APPENDIX A6 · RANKING MODEL

SOFT score: $\Sigma w_i \cdot S_i$ — 0.35 of total weight comes from signals only Tasco can generate

7 components, sum to 1.00. Missing data → drop the component + renormalize, never fake a 0.5.

DEFAULT WEIGHTS (untuned — say so plainly to judges)



S_route + S_behavior (+ S_crowd) = 0.35 — map/infrastructure signals

a competitor copying the formula still has no input · share only grows as real VETC data lands

FORMULAS WORTH REMEMBERING

`S_route = exp(-detour_min / 5)`

0' = 1.0 · 5' ≈ 0.37 · smooth decay, no hard cliff

`S_crowd = 1 - occupancy(t_arrival)/100`

RENORMALIZE (missing data)

`w_i' = w_i / Σ w_with_data`

15-query benchmark: S_behavior + every SIM signal is zeroed

→ same mechanism handles cold-start in production

MODE SWITCH (Meituan RecSys'24: repeat and explore need different recommenders)

repeat — familiar route (≥3 trips/90d)

semantic 0.25 · behavior 0.25 · buzz 0.00

"your usual place on this route" — behavior is king

explore — holiday / new route

semantic 0.35 · behavior 0.05 · buzz 0.10

regional specialties + what's trending surface first

S_match is multiplied by **confidence from the resolver**: parking confirmed by a driver (0.85) contributes more than an agentic guess (0.60) — ranking **inherits the cascade's trust**. No other Vietnamese system does this.